Selective Search for Object Recognition

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Abstract and Figures

This paper addresses the problem of generating possible object locations for use in object recognition. We introduce selective search which combines the strength of both an exhaustive search and segmentation. Like segmentation, we use the image structure to guide our sampling process. Like exhaustive search, we aim to capture all possible object locations. Instead of a single technique to generate possible object locations, we diversify our search and use a variety of complementary image partitionings to deal with as many image conditions as possible. Our selective search results in a small set of data-driven, class-independent, high quality locations, yielding 99 % recall and a Mean Average Best Overlap of 0.879 at 10,097 locations. The reduced number of locations compared to an exhaustive search enables the use of stronger machine learning techniques and stronger appearance models for object recognition. In this paper we show that our selective search enables the use of the powerful Bag-of-Words model for recognition. The selective search software is made publicly available (Software: http://disi.unitn.it/~uijlings/SelectiveSearch.html).

Two examples of our selective…

Our selective search method…

The Average Best Overlap scores…

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Selective Search for Object Recognition

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Abstract

This paper addresses the problem of generating possible object locations for use in object recognition. We introduce Selective Search which combines the strength of both an exhaustive search and segmentation. Like segmentation, we use the image structure to guide our sampling process. Like exhaustive search, we aim to capture all possible object locations. Instead of a single technique to generate possible object locations, we *diversify* our search and use a variety of complementary image partitionings to deal with as many image conditions as possible. Our Selective Search results in a small set of data-driven, class-independent, high quality locations, yielding 99% recall and a Mean Average Best Overlap of 0.879 at 10,097 locations. The reduced number of locations compared to an exhaustive search enables the use of stronger machine learning techniques and stronger appearance models for object recognition. In this paper we show that our selective search enables the use of the powerful Bag-of-Words model for recognition. The Selective Search software is made publicly available [1].

1 Introduction

For a long time, objects were sought to be delineated before their identification. This gave rise to segmentation, which aims for a unique partitioning of the image through a generic algorithm, where there is one part for all object silhouettes in the image. Research on this topic has yielded tremendous progress over the past years [3, 6, 13, 26]. But images are intrinsically hierarchical: In

(a) (b)

(c) (d)

Figure 1: There is a high variety of reasons that an imag forms an object. In (b) the cats can be distinguished by co texture. In (c) the chameleon can be distinguished from rounding leaves by texture, not colour. In (d) the wheels ca of the car because they are enclosed, not because they ar in texture or colour. Therefore, to find objects in a structu it is necessary to use a variety of diverse strategies. Furt an image is intrinsically hierarchical as there is no single which the complete table, salad bowl, and salad spoon can in (a).

Figure 1a the salad and spoons are inside the salad bowl, which in turn stands on the table. Furthermore, depending on the context the term *table* in this picture can refer to only the wood or include everything on the table. Therefore both the nature of images and the different uses of an object category are hierarchical. This prohibits the unique partitioning of objects for all but the most specific purposes. Hence for most tasks multiple scales in a segmentation are a necessity. This is most naturally addressed by using a hierarchical partitioning, as done for example by Arbelaez *et al.* [3].

Besides that a segmentation should be hierarchical, a generic solution for segmentation using a single strategy may not exist at all. There are many conflicting reasons why a region should be grouped together: In Figure 1b the cats can be separated using colour, but their texture is the same. Conversely, in Figure 1c the chameleon

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is similar to its surrounding leaves in terms of colour, ye ture differs. Finally, in Figure 1d, the wheels are wildly from the car in terms of both colour and texture, yet are by the car. Individual visual features therefore cannot re ambiguity of segmentation.

And, finally, there is a more fundamental problem. Reg very different characteristics, such as a face over a swe only be combined into one object after it has been establi the object at hand is a human. Hence without prior recogn hard to decide that a face and a sweater are part of one ob

This has led to the opposite of the traditional approac localisation through the identification of an object. This r proach in object recognition has made enormous progre than a decade [8, 12, 16, 35]. With an appearance mode from examples, an exhaustive search is performed where cation within the image is examined as to not miss any object location [8, 12, 16, 35].

However, the exhaustive search itself has several drawbacks. Searching every possible location is computationally infeasible. The search space has to be reduced by using a regular grid, fixed scales, and fixed aspect ratios. In most cases the number of locations to visit remains huge, so much that alternative restrictions need to be imposed. The classifier is simplified and the appearance model needs to be fast. Furthermore, a uniform sampling yields many boxes for which it is immediately clear that they are not supportive of an object. Rather then sampling locations blindly using an exhaustive search, a key question is: Can we steer the sampling by a data-driven analysis?

In this paper, we aim to combine the best of the intuitions of segmentation and exhaustive search and propose a data-driven *selective search*. Inspired by bottom-up segmentation, we aim to exploit the structure of the image to generate object locations. Inspired by exhaustive search, we aim to capture all possible object locations. Therefore, instead of using a single sampling technique, we aim to *diversify* the sampling techniques to account for as many image conditions as possible. Specifically, we use a data-driven grouping-based strategy where we increase diversity by using a variety of complementary grouping criteria and a variety of complementary colour spaces with different invariance properties. The set of locations is obtained by combining the locations of these complementary partitionings. Our goal is to generate a class-independent, data-driven, selective search strategy that generates a small set of high-quality object locations.

Our application domain of selective search is object recognition. We therefore evaluate on the most commonly used dataset for this purpose, the Pascal VOC detection challenge which consists of 20 object classes. The size of this dataset yields computational constraints for our selective search. Furthermore, the use of this dataset means that the quality of locations is mainly evaluated in terms of bounding boxes. However, our selective search applies to regions as well and is also applicable to concepts such as "grass".

In this paper we propose selective search for object recognition. Our main research questions are: (1) What are good diversification strategies for adapting segmentation as a selective search strategy? (2) How effective is selective search in creating a small set of high-

such as HOG [8, 16, 36]. This method is often used as a tion step in a cascade of classifiers [16, 36].

Related to the sliding window technique is the highly s part-based object localisation method of Felzenszwalb *et* Their method also performs an exhaustive search using SVM and HOG features. However, they search for ob object parts, whose combination results in an impressive c tection performance.

Lampert *et al.* [17] proposed using the appearance r guide the search. This both alleviates the constraints o regular grid, fixed scales, and fixed aspect ratio, while at time reduces the number of locations visited. This is do rectly searching for the optimal window within the imag branch and bound technique. While they obtain impressiv for linear classifiers, [1] found that for non-linear class method in practice still visits over a 100,000 windows pe

Instead of a blind exhaustive search or a branch ar search, we propose selective search. We use the underl age structure to generate object locations. In contrast to cussed methods, this yields a completely class-independe locations. Furthermore, because we do not use a fixed a tio, our method is not limited to objects but should be ab stuff like "grass" and "sand" as well (this also holds for [nally, we hope to generate fewer locations, which should problem easier as the variability of samples becomes lov more importantly, it frees up computational power whic used for stronger machine learning techniques and more appearance models.

2.2 Segmentation

Both Carreira and Sminchisescu [4] and Endres and Hoien pose to generate a set of class independent object hypothe segmentation. Both methods generate multiple foregrou ground segmentations, learn to predict the likelihood tha ground segment is a complete object, and use this to rank ments. Both algorithms show a promising ability to a delineate objects within images, confirmed by [19] wh quality locations within an image? (3) Can we use selective search to employ more powerful classifiers and appearance models for object recognition?

2 Related Work

state-of-the-art results on pixel-wise image classification u
As common in segmentation, both methods rely on a sing
algorithm for identifying good regions. They obtain a v
locations by using many randomly initialised foreground a
ground seeds. In contrast, we explicitly deal with a variety
conditions by using different grouping criteria and differe
sentations. This means a lower computational investment

As an object can be located at any position and scale in the image, it is natural to search everywhere [8, 16, 36]. However, the visual search space is huge, making an exhaustive search computationally expensive. This imposes constraints on the evaluation cost per location and/or the number of locations considered. Hence most of these sliding window techniques use a coarse search grid and fixed aspect ratios, using weak classifiers and economic image features

recognizing objects based on their parts. They first gene
of part hypotheses using a grouping method based on Ar
al. [3]. Each part hypothesis is described by both appear
shape features. Then, an object is recognized and careful
eated by using its parts, achieving good results for shape
tion. In their work, the segmentation is hierarchical and yi
ments at all scales. However, they use a single grouping

Figure 2: Two examples of our selective search showing the necessity of different scales. On the left we find many objects at scales. On the right we necessarily find the objects at different scales as the girl is contained by the tv.

whose power of discovering parts or objects is left unevaluated. In this work, we use multiple complementary strategies to deal with as many image conditions as possible. We include the locations generated using [3] in our evaluation.

2.3 Other Sampling Strategies

Alexe *et al.* [2] address the problem of the large sampling space of an exhaustive search by proposing to search for *any* object, independent of its class. In their method they train a classifier on the

3 Selective Search

In this section we detail our selective search algorithm f
recognition and present a variety of diversification strategi
with as many image conditions as possible. A selective s
gorithm is subject to the following design considerations:

Capture All Scales. Objects can occur at any scale withi
age. Furthermore, some objects have less clear bo
then other objects. Therefore, in selective search a
scales have to be taken into account, as illustrated i
2. This is most naturally achieved by using an hie
algorithm.

object windows of those objects which have a well-defined shape (as opposed to stuff like "grass" and "sand"). Then instead of a full exhaustive search they randomly sample boxes to which they apply their classifier. The boxes with the highest "objectness" measure serve as a set of object hypotheses. This set is then used to greatly reduce the number of windows evaluated by class-specific object detectors. We compare our method with their work.

Another strategy is to use visual words of the Bag-of-Words model to predict the object location. Vedaldi *et al.* [34] use jumping windows [5], in which the relation between individual visual words and the object location is learned to predict the object location in new images. Maji and Malik [23] combine multiple of these relations to predict the object location using a Hough-transform, after which they randomly sample windows close to the Hough maximum. In contrast to learning, we use the image structure to sample a set of class-independent object hypotheses.

To summarize, our novelty is as follows. Instead of an exhaustive search [8, 12, 16, 36] we use segmentation as selective search yielding a small set of class independent object locations. In contrast to the segmentation of [4, 9], instead of focusing on the best segmentation algorithm [3], we use a variety of strategies to deal with as many image conditions as possible, thereby severely reducing computational costs while potentially capturing more objects accurately. Instead of learning an objectness measure on randomly sampled boxes [2], we use a bottom-up grouping procedure to generate good object locations.

Diversification. There is no single optimal strategy to g gions together. As observed earlier in Figure 1, reg form an object because of only colour, only texture, o parts are enclosed. Furthermore, lighting condition shading and the colour of the light may influence hov form an object. Therefore instead of a single strateg works well in most cases, we want to have a diver strategies to deal with all cases.

Fast to Compute. The goal of selective search is to yiel possible object locations for use in a practical object tion framework. The creation of this set should not b computational bottleneck, hence our algorithm shoul sonably fast.

3.1 Selective Search by Hierarchical Group

We take a hierarchical grouping algorithm to form the bas selective search. Bottom-up grouping is a popular approac mentation [6, 13], hence we adapt it for selective search. the process of grouping itself is hierarchical, we can natur erate locations at all scales by continuing the grouping pro the whole image becomes a single region. This satisfies th tion of capturing all scales.

As regions can yield richer information than pixels, we use region-based features whenever possible. To get a set starting regions which ideally do not span multiple object

3

the fast method of Felzenszwalb and Huttenlocher [13], which [3] found well-suited for such purpose.

Our grouping procedure now works as follows. We first use [13] to create initial regions. Then we use a greedy algorithm to iteratively group regions together: First the similarities between all neighbouring regions are calculated. The two most similar regions are grouped together, and new similarities are calculated between the resulting region and its neighbours. The process of grouping the most similar regions is repeated until the whole image becomes a single region. The general method is detailed in Algorithm 1.

Algorithm 1: Hierarchical Grouping Algorithm

Input: (colour) image
Output: Set of object location hypotheses L

Obtain initial regions $R = \{r_1, \cdots, r_n\}$ using [13]
Initialise similarity set $S = \emptyset$
foreach *Neighbouring region pair* (r_i, r_j) **do**
  Calculate similarity $s(r_i, r_j)$
  $S = S \cup s(r_i, r_j)$

while $S \neq \emptyset$ **do**
  Get highest similarity $s(r_i, r_j) = \max(S)$
  Merge corresponding regions $r_t = r_i \cup r_j$
  Remove similarities regarding $r_i : S = S \setminus s(r_i, r_*)$
  Remove similarities regarding $r_j : S = S \setminus s(r_*, r_j)$
  Calculate similarity set S_t between r_t and its neighbours
  $S = S \cup S_t$
  $R = R \cup r_t$

colour channels	R	G	B	I	V	L	a	b	S	r	g
Light Intensity	-	-	-	-	-	-	+/-	+/-	+	+	+
Shadows/shading	-	-	-	-	-	-	+/-	+/-	+	+	+
Highlights	-	-	-	-	-	-	-	-	-	-	-

colour spaces	RGB	I	Lab	rgI	HSV	rgb	
Light Intensity	-	-	+/-	$2/3$	$2/3$	+	
Shadows/shading	-	-	+/-	$2/3$	$2/3$	+	
Highlights	-	-	-	-	$1/3$	-	-

Table 1: The invariance properties of both the individu channels and the colour spaces used in this paper, sorte gree of invariance. A "+/-" means partial invariance. A $1/3$ means that one of the three colour channels is invaria property.

these images we rely on the other diversification methoc suring good object locations.

In this paper we always use a single colour space th the algorithm, meaning that both the initial grouping algo [13] and our subsequent grouping algorithm are performe colour space.

Complementary Similarity Measures. We define four mentary, fast-to-compute similarity measures. These mea all in range $[0, 1]$ which facilitates combinations of these r

$s_{colour}(r_i, r_j)$ measures colour similarity. Specifically, for gion we obtain one-dimensional colour histograms

Extract object location boxes L from all regions in R

For the similarity $s(r_i, r_j)$ between region r_i and r_j we want a variety of complementary measures under the constraint that they are fast to compute. In effect, this means that the similarities should be based on features that can be propagated through the hierarchy, *i.e.* when merging region r_i and r_j into r_t, the features of region r_t need to be calculated from the features of r_i and r_j without accessing the image pixels.

3.2 Diversification Strategies

The second design criterion for selective search is to diversify the sampling and create a set of complementary strategies whose locations are combined afterwards. We diversify our selective search (1) by using a variety of colour spaces with different invariance properties, (2) by using different similarity measures s_{ij}, and (3) by varying our starting regions.

Complementary Colour Spaces. We want to account for different scene and lighting conditions. Therefore we perform our hierarchical grouping algorithm in a variety of colour spaces with a range of invariance properties. Specifically, we the following colour spaces with an increasing degree of invariance: (1) RGB, (2) the intensity (grey-scale image) I, (3) Lab, (4) the rg channels of normalized RGB plus intensity denoted as rgI, (5) HSV, (6) normalized RGB denoted as rgb, (7) C [14] which is an opponent colour space where intensity is divided out, and finally (8) the Hue channel H from HSV. The specific invariance properties are listed in Table 1.

Of course, for images that are black and white a change of colour space has little impact on the final outcome of the algorithm. For

This leads to a colour histogram $C_i = \{c_i^1, \cdots, c_i^n\}$ for each region r_i with dimensionality $n = 75$ when three colo
nels are used. The colour histograms are normalised
L_1 norm. Similarity is measured using the histogram
tion:

$$s_{colour}(r_i, r_j) = \sum_{k=1}^{n} \min(c_i^k, c_j^k).$$

The colour histograms can be efficiently propagated
the hierarchy by

$$C_t = \frac{\text{size}(r_i) \times C_i + \text{size}(r_j) \times C_j}{\text{size}(r_i) + \text{size}(r_j)}.$$

The size of a resulting region is simply the sum of
stituents: $\text{size}(r_t) = \text{size}(r_i) + \text{size}(r_j)$.

$s_{texture}(r_i, r_j)$ measures texture similarity. We represent te
ing fast SIFT-like measurements as SIFT itself work
material recognition [20]. We take Gaussian deriv
eight orientations using $\sigma = 1$ for each colour chan
each orientation for each colour channel we extra
togram using a bin size of 10. This leads to a tex
togram $T_i = \{t_i^1, \cdots, t_i^n\}$ for each region r_i with di
ality $n = 240$ when three colour channels are used.
histograms are normalised using the L_1 norm. Sim
measured using histogram intersection:

$$s_{texture}(r_i, r_j) = \sum_{k=1}^{n} \min(t_i^k, t_j^k).$$

Texture histograms are efficiently propagated throug
erarchy in the same way as the colour histograms.

$s_{size}(r_i, r_j)$ encourages small regions to merge early. This forces regions in S, *i.e.* regions which have not yet been merged, to be of similar sizes throughout the algorithm. This is desirable because it ensures that object locations at all scales are created at all parts of the image. For example, it prevents a single region from gobbling up all other regions one by one, yielding all scales only at the location of this growing region and nowhere else. $s_{size}(r_i, r_j)$ is defined as the fraction of the image that r_i and r_j jointly occupy:

$$s_{size}(r_i, r_j) = 1 - \frac{\text{size}(r_i) + \text{size}(r_j)}{\text{size}(im)}, \quad (4)$$

where $\text{size}(im)$ denotes the size of the image in pixels.

$s_{fill}(r_i, r_j)$ measures how well region r_i and r_j fit into each other. The idea is to fill gaps: if r_i is contained in r_j it is logical to merge these first in order to avoid any holes. On the other hand, if r_i and r_j are hardly touching each other they will likely form a strange region and should not be merged. To keep the measure fast, we use only the size of the regions and of the containing boxes. Specifically, we define BB_{ij} to be the tight bounding box around r_i and r_j. Now $s_{fill}(r_i, r_j)$ is the fraction of the image contained in BB_{ij} which is not covered

object hypothesis set, depending on the computational effi
the subsequent feature extraction and classification metho

We choose to order the combined object hypotheses
on the order in which the hypotheses were generated in
dividual grouping strategy. However, as we combine resu
up to 80 different strategies, such order would too heavil
size large regions. To prevent this, we include some ran
as follows. Given a grouping strategy j, let r_i^j be the regi
is created at position i in the hierarchy, where $i = 1$ repre
top of the hierarchy (whose corresponding region covers
plete image). We now calculate the position value v_i^j as
where RND is a random number in range $[0, 1]$. The fina
is obtained by ordering the regions using v_i^j.

When we use locations in terms of bounding boxes, we
all the locations as detailed above. Only afterwards we
lower ranked duplicates. This ensures that duplicate box
better chance of obtaining a high rank. This is desirable
if multiple grouping strategies suggest the same box loca
likely to come from a visually coherent part of the image.

4 Object Recognition using Sel Search

r_i and r_j:

$$fill(r_i, r_j) = 1 - \frac{\text{size}(BB_{ij}) - \text{size}(r_i) - \text{size}(r_i)}{\text{size}(im)} \quad (5)$$

We divide by size(im) for consistency with Equation 4. Note that this measure can be efficiently calculated by keeping track of the bounding boxes around each region, as the bounding box around two regions can be easily derived from these.

In this paper, our final similarity measure is a combination of the above four:

$$\begin{aligned} s(r_i, r_j) &= a_1 s_{colour}(r_i, r_j) + a_2 s_{texture}(r_i, r_j) + \\ &\quad a_3 s_{size}(r_i, r_j) + a_4 s_{fill}(r_i, r_j), \end{aligned} \quad (6)$$

where $a_i \in \{0, 1\}$ denotes if the similarity measure is used or not. As we aim to diversify our strategies, we do not consider any weighted similarities.

Complementary Starting Regions. A third diversification strategy is varying the complementary starting regions. To the best of our knowledge, the method of [13] is the fastest, publicly available algorithm that yields high quality starting locations. We could not find any other algorithm with similar computational efficiency so we use only this oversegmentation in this paper. But note that different starting regions are (already) obtained by varying the colour spaces, each which has different invariance properties. Additionally, we vary the threshold parameter k in [13].

3.3 Combining Locations

In this paper, we combine the object hypotheses of several variations of our hierarchical grouping algorithm. Ideally, we want to order the object hypotheses in such a way that the locations which are most likely to be an object come first. This enables one to find a good trade-off between the quality and quantity of the resulting

This paper uses the locations generated by our selective s object recognition. This section details our framework f recognition.

Two types of features are dominant in object recognit tograms of oriented gradients (HOG) [8] and bag-of-word HOG has been shown to be successful in combination with based model by Felzenszwalb *et al.* [12]. However, as th exhaustive search, HOG features in combination with a lin sifier is the only feasible choice from a computational per In contrast, our selective search enables the use of more e and potentially more powerful features. Therefore we us words for object recognition [16, 17, 34]. However, we us powerful (and expensive) implementation than [16, 17, 34 ploying a variety of colour-SIFT descriptors [32] and a fin pyramid division [18].

Specifically we sample descriptors at each pixel on a sin ($\sigma = 1.2$). Using software from [32], we extract SIFT [21 colour SIFTs which were found to be the most sensitiv tecting image structures, Extended OpponentSIFT [31] a SIFT [32]. We use a visual codebook of size 4,000 and pyramid with 4 levels using a 1x1, 2x2, 3x3. and 4x4 This gives a total feature vector length of 360,000. In im sification, features of this size are already used [25, 37]. a spatial pyramid results in a coarser spatial subdivision cells which make up a HOG descriptor, our features con information about the specific spatial layout of the objec fore, HOG is better suited for rigid objects and our fea better suited for deformable object types.

As classifier we employ a Support Vector Machine w togram intersection kernel using the Shogun Toolbox [28 ply the trained classifier, we use the fast, approximate clas strategy of [22], which was shown to work well for Bag- in [30].

Our training procedure is illustrated in Figure 3. The ini tive examples consist of all ground truth object windows. negative examples we select from all object locations g

... Real-time vehicle detection is a basis for every current intelligent transport system. Traditional vehicle detection techniques include background subtraction, edge detection, and motion tracking, yet they fall short of accuracy because of the dynamic conditions induced by occlusion, varying illumination, or high traffic density [14] . Convolutional neural networks (CNN) have completely transformed the arena of fast and accurate real-time vehicle detection [15]. ...

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... Object detection is more challenging than image classification because it involves recognising and localising objects in an image [10]. Early object detection models, such as region-based CNN (R-CNN) [11] used region proposal algorithms such as selective search [45] to propose regions of interest in an image, extract features from these regions using a convolutional neural network (CNN) [34], and classify the objects and wrap them in a bounding box. Region proposals were identified as a bottleneck, which the successors of R-CNN attempted to improve on [10,36]. ...

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